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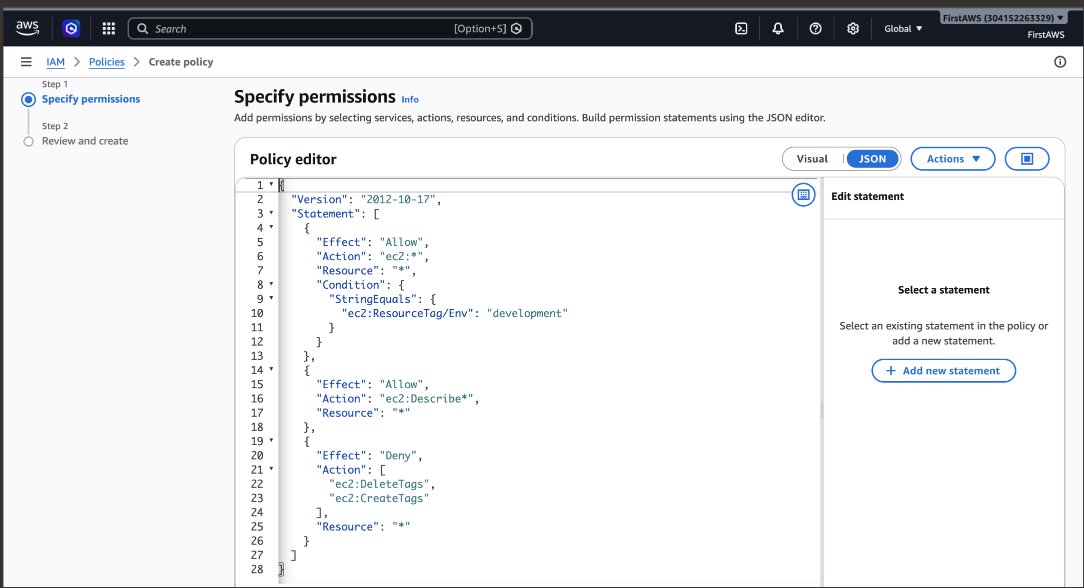
Cloud Security with AWS IAM

JJ jjoseph1608@outlook.com



Introducing Today's Project!

Project overview

Today we are using AWS IAM to control access to cloud resources. We will launch an EC2 instance and then create IAM policies and user groups to manage who is authenticated and authorised to access it.

Tools and concepts

In this project I learned about Amazon EC2, which allows you to rent virtual computers in the cloud, and launched two instances tagged for production and development environments. I learned about AWS IAM, including how to create IAM policies to control access to resources, user groups to manage permissions for multiple users at once, and IAM users to give individuals their own login credentials. I also created an account alias to make the sign-in URL more user friendly, and tested the intern's permissions to confirm the policy was working correctly.

Project reflection

This project took me approximately 1 hour to complete. The most challenging part was understanding and writing the IAM policy in JSON, as it required careful attention to the structure and conditions to ensure the right permissions were applied. It was most rewarding to see the policy working correctly in practice, attempting to stop the production instance and seeing it fail, then successfully stopping the development instance, which really brought the whole concept to life.

Tags

What I did in this step

In this step, I will be launching two amazon EC2 instances so that we can increase computing power

Understanding tags

Tags are like labels you can attach to AWS resources for organization.

My tag configuration

The tag I've used on my EC2 instances is called Env. The value I've assigned for my instances are production and development

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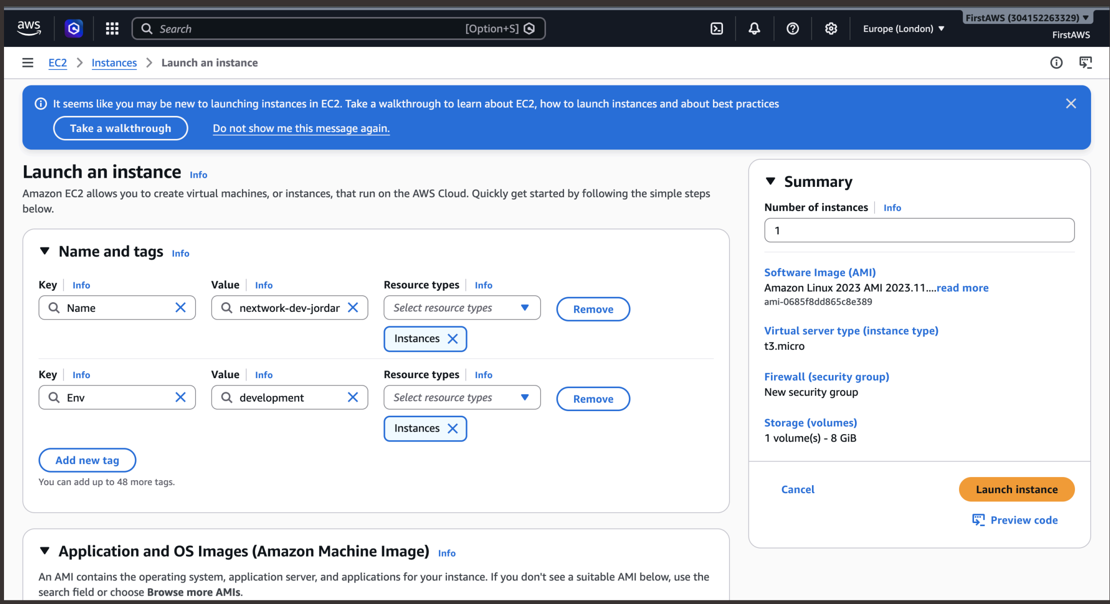


IAM Policies

What I did in this step

In this step, I will create an IAM policy that gives access to the developement instance.

Understanding IAM policies

An IAM policy is a rule for who can do what with your AWS resources. It's all about giving permissions to IAM users, groups, or roles, saying what they can or can't do on certain resources, and when those rules kick in.

The policy I set up

For this project, I’ve set up a policy using JSON

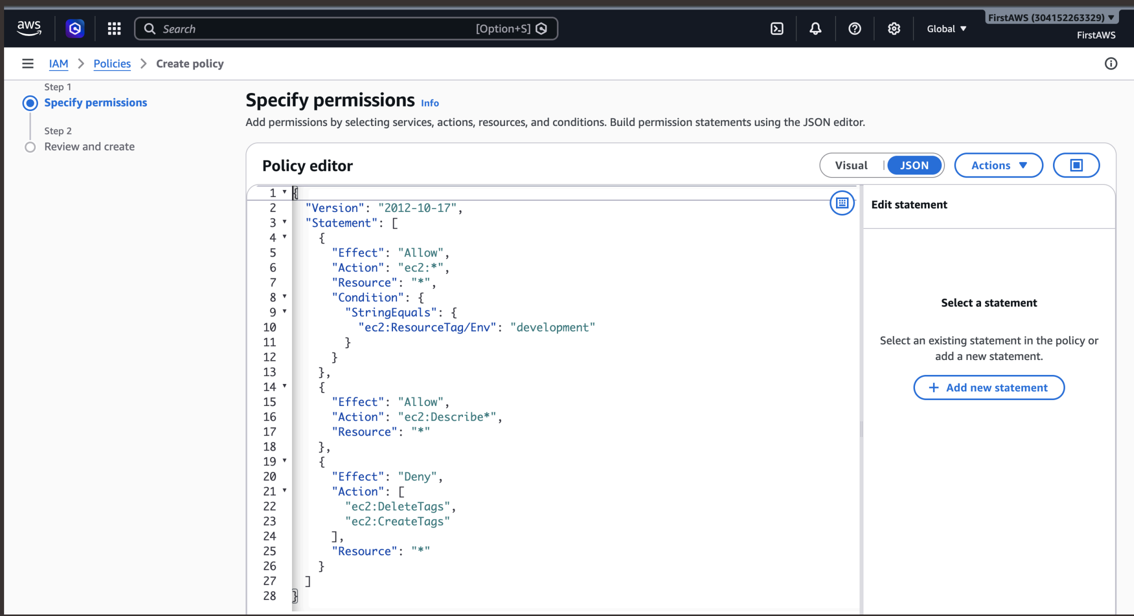
Policy effect

This policy allows some actions (like starting, stopping, and describing EC2 instances) for instances tagged with "Env = development" while denying the ability to create or delete tags for all instances.

Understanding Effect, Action, and Resource

Effect This can have two values - either Allow or Deny - to indicate whether the policy allows or denies a certain action. Deny has priority. Looking at the first statement, "Effect": "Allow" means this statement is trying to allow for an action. Action A list of the actions that the policy allows or denies. In this case, "Action": "ec2:*" means all actions that you could possibly take on EC2 instances are allowed. Woohoo! Resource Which resources does this policy apply to? Specifying "*" means all resources within the defined scope (see the next point).

My JSON Policy



Account Alias

What I did in this step

In this step, I will simplify user login to your AWS account using an Account Alias.

Understanding account aliases

An Account Alias is a friendly, readable name you can create for your AWS account to replace the default account ID, which is usually a long string of numbers. Instead of sharing a complex URL like https://123456789.signin.aws.amazon.com/console, you can share something much simpler like https://nextwork-alias.signin.aws.amazon.com/console, making it easier for new users like the intern to find and log in to the AWS account.

Setting up my account alias

Creating an account alias took me just a few seconds. Now, my new AWS console sign-in URL is https://nextwork-alias-jordan-joseph.signin.aws.amazon.com/console.

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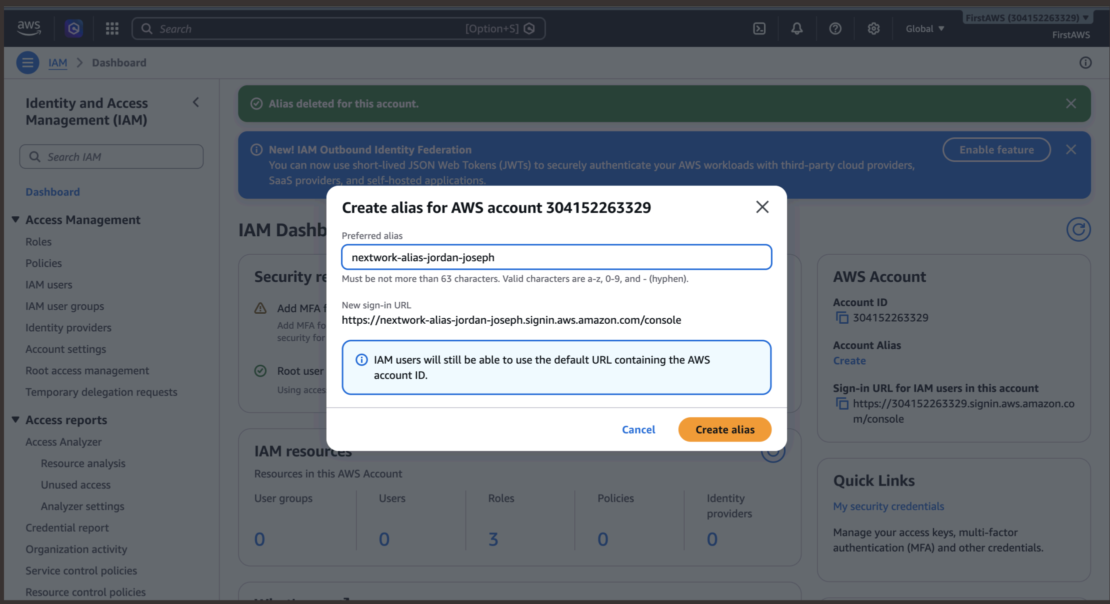


IAM Users and User Groups

What I did in this step

In this step, I will set up a dedicated IAM group for all NextWork interns, so I can manage all interns' permissions from one place and set up a dedicated IAM user for the new intern, so they have a way to log in.

Understanding user groups

An IAM user group is a collection/folder of IAM users. It allows you to manage permissions for all the users in your group at the same time by attaching policies to the group rather than individual users.

Attaching policies to user groups

I attached the policy I created to this user group, which means that the user policy applies to all that are in the user group

Understanding IAM users

IAM users are the people that will get access to my resources/AWS account, whereas user groups are the collections/folders of users for easier user management.

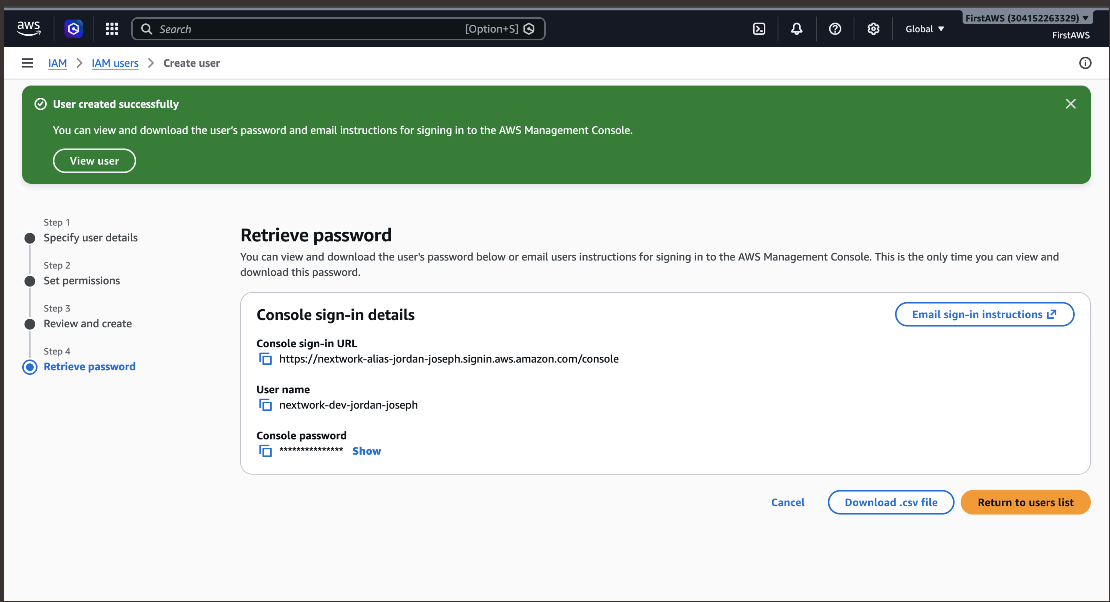
Logging in as an IAM User

Sharing sign-in details

You can share a new user's sign-in details by either copying the console sign-in URL along with their username and password, or by downloading a CSV file containing all of their credentials.

Observations from the IAM user dashboard

Once I logged in as my IAM user, I noticed that some of the dashboard panels were showing up as access denied due to the user group policy associated with the user



Testing IAM Policies

What I did in this step

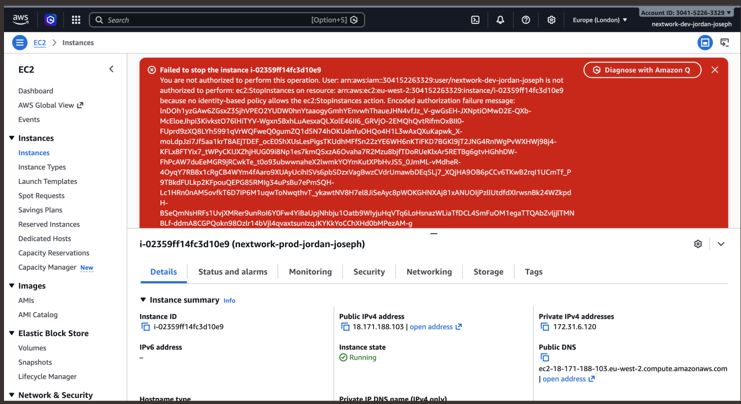
In this step, I will log into AWS using the intern's IAM user and test the access to the production and development instance.

Testing policy actions

I stopped both EC2 instances — first attempting to stop the production instance, which failed due to insufficient permissions, and then successfully stopping the development instance, confirming that the IAM policy was working correctly.

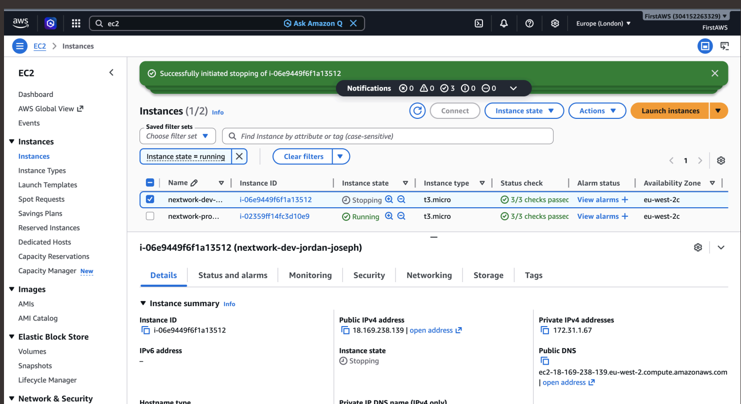
Stopping the production instance

When I tried to stop the production instance I received an error. This was due to not having the right permissions.



Stopping the development instance

It was successful. The development instance stopped without any issues, confirming that the IAM user had the correct permissions to manage instances tagged with the development environment.



IAM Policy Simulator

I will be using a special IAM tool called the Policy Simulator to validate policies without affecting your actual AWS resources.

Understanding the IAM Policy Simulator

I would use a policy simulator to validate policies without affecting my actual AWS resources

How I used the simulator

The simulation results for the development instance showed that the actions were allowed, confirming that the IAM user had the correct permissions to manage and interact with the development environment.

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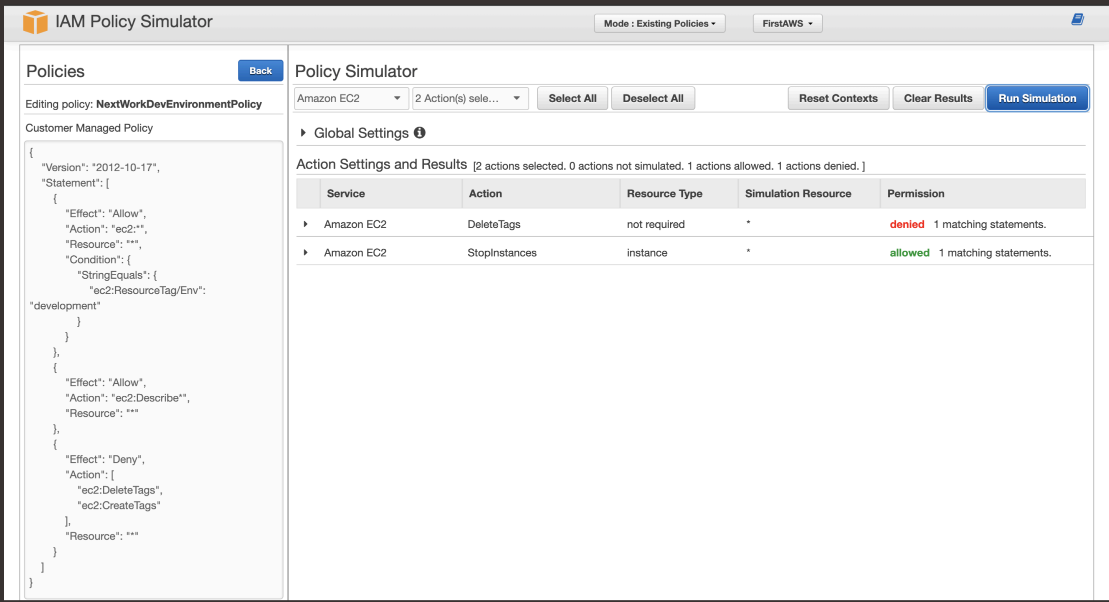

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